# CSR-2315 Data Structures

# Week 3: Worst-Case Run Time Analysis

Emmanuel Garcia

### 1. is_even(N)

**Answer: O (1)**

**Reasoning:** The function performs a ***fixed*** number of operations no matter the value of N; N is either divisible by 2 or not.

### 2. find_number(arr, number)

**Answer: O(N)**

**Reasoning:** In worst case scenario, the function must check every element to ensure that the same value isn't present.

### 3. constant_multiplier(arr)

**Answer: O(N)**

**Reasoning:** The loop goes through every element in the array once. Since there are N elements, the number of operations grows linearly with N.

### 4. arr = [2*elem for elem in arr]

**Answer: O(N)**

**Reasoning:** The entire array is gone through once; the more items present the longer it will take.

### 5. create_pairs(arr)

**Answer: O(N^2)**

**Reasoning:** The function contains a loop inside another loop. The amount of time increases as much as N^2.

### 6. is_prime(N)

**Answer: O(N)**

**Reasoning:** The loop checks every integer from 2 up to N. Therefore, the number of operations grows linearly with N, which increases time.

### 7. print_halves(N)

**Answer: O(log N)**

**Reasoning:** Each iteration divides N by 2 (N/2). The number of times N is divided by 2 the slower the results, i.e. 12 loops.

### 8. double_or_nothing(arr)

**Answer: O(N)**

**Reasoning:** Two-step process, each processing the elements once.

### 9. Determine the Big-O notation from the exact operation count.

A. N/2

Answer: O(N)

Reasoning: Dividing N by a constant does not change its linear growth; O(N).

### B. 3N + 5

**Answer: O(N)**

**Reasoning:** Constants 3 and 5 are being ignored in Big-O notation. The highest-order term is N, so the run time is O(N).

### C. 36,000

**Answer: O (1)**

**Reasoning:** The number of operations is constant and not depending on N.

**D. N^3 + 2N^2 + N + 3000**

**Answer: O(N^3)**

**Reasoning:** The highest-order term is N^3; O(N^3)

E. 1000N + (1/1000)N²

Answer: O(N^2)

Reasoning: N² grows faster than N, so the quadratic term dominates; O(N²).

**F. N × (N − 1) × (N − 2) × (N − 3) × ... × 1**

**Answer: O(N!)**

**Reasoning:** The expression is N factorial; therefore, its run time is O(N!).

**G. Σ(N − i), from i = 0 to N**

**Answer: O(N²)**

**Reasoning:** The sum produces approximately N^2 / 2 operations, giving O(N²).